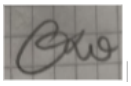
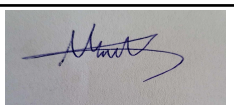
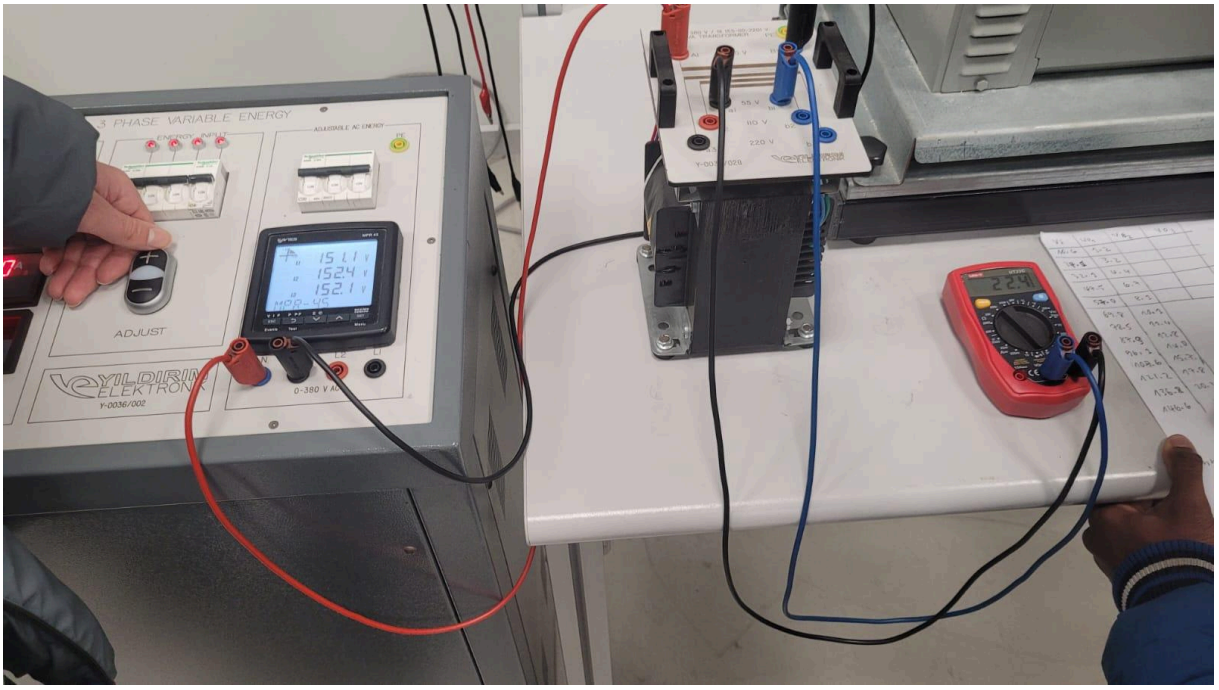
EE3014 Energy Conversion

Experiment 20

Student No	Name	Surname	Contribution	Signature
150722923	Jacques	Asinyo	Abstract, Introduction, theoretical background	
150725824	Kadir Baha	Demirhan	Results, Conclusion, Questions	
150720062	Mehmet Ali	KAMALAK	Procedure and experimental setup	
150722066	Mevlüt	Akhan	Experimental Setup, Procedure and Graphs	



1. Objective

The objective of this experiment is to investigate the relationship between the input primary voltage (Vi) and the output secondary voltages (Vo) of a single-phase transformer. Specifically, the experiment aims to:

- Verify the transformer turns ratio relationship.
- Analyze the linearity of the transformer across a wide range of input voltages (from approx. 10V to 220V).
- Observe deviations in the voltage ratio (Vo / Vi) at different operating points.

2. Equipment Used

- **Transformer Module:** Single phase transformer (Model Y-036/027).
- **Power Supply:** Variable AC Power Source (0 - 250V)
- **Measurement Tools:** Digital Multimeters (Voltmeter) for measuring Vi and Vo.

3. Theory

3.1 Ideal Transformer Principles

An ideal single-phase transformer operates on the principle of electromagnetic induction as defined by **Faraday's Law**. When an alternating voltage is applied to the primary winding, it creates a time-varying magnetic flux (Φ) in the core. This flux links

both the primary and secondary windings, inducing an electromotive force (EMF) in each.

According to Faraday's Law, the induced voltage (E) in a coil is proportional to the number of turns (N) and the rate of change of magnetic flux (dΦ/dt):

$E = -N * (d\Phi/dt)$

Since the same magnetic flux links both windings in an ideal transformer, the voltage ratio is directly determined by the turns ratio:

$Vi / Vo = Np / Ns = a$

Where:

- **Vi** and **Vo** are the Primary and Secondary voltages.
- **Np** and **Ns** are the number of turns in the primary and secondary windings.
- **a** is the transformation ratio.

In this ideal model, the transformer has no internal resistance, perfect magnetic coupling (no leakage), and infinite core permeability. Consequently, the ratio Vo/Vi would be constant for all input voltages.

3.2 Real Transformer Behavior (Non-Ideal)

In practical experiments, the measured voltage ratio deviates from the theoretical ideal due to physical limitations of the materials and geometry.

A. Winding Resistance and Voltage Drop

The copper wire used for windings has a finite internal resistance (R1 for primary, R2 for secondary). Even when there is no load connected to the output, a small current flows into the primary winding to magnetize the core (Excitation Current).

- This current flows through the primary resistance, causing a voltage drop according to Ohm’s law: **V_drop = I_exc * R1**.
- As a result, the actual induced voltage (E1) is slightly less than the applied input voltage (Vi): **E1 = Vi - (I_exc * R1)**.
- This effect is most noticeable at very low input voltages, where the resistive drop is a significant fraction of the total applied voltage, causing the Vo/Vi ratio to appear lower than expected.

B. Magnetic Leakage Flux

In a real transformer, not all magnetic flux produced by the primary winding reaches the secondary winding. Some flux "leaks" through the air surrounding the coils.

- This is modeled as **Leakage Reactance (X)** in the equivalent circuit.
- Leakage flux results in a reactive voltage drop that further reduces the output voltage, especially as current increases.

C. Core Saturation and Non-Linearity

The magnetic core (usually made of silicon steel) does not behave linearly forever. As the input voltage increases, the magnetic flux density (B) in the core increases.

- **Linear Region:** At moderate voltages, the core's permeability is roughly constant, and the voltage ratio is stable.
- **Saturation Region:** At very high voltages, the core reaches **Magnetic Saturation**. The material cannot support a further proportional increase in flux. This leads to a sharp increase in magnetizing current and a distortion of the voltage waveform, causing the linear relationship between Vi and Vo to break down.

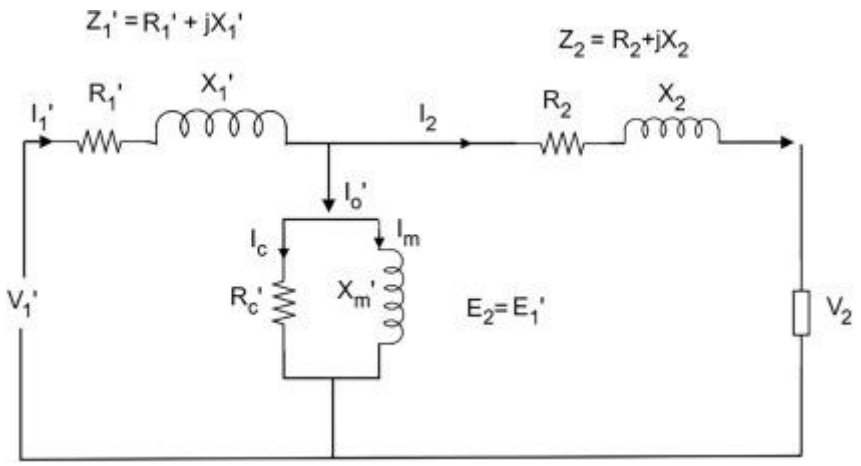


Figure 1: Equivalent circuit of a real transformer showing internal resistance and leakage reactance.

4. Procedure

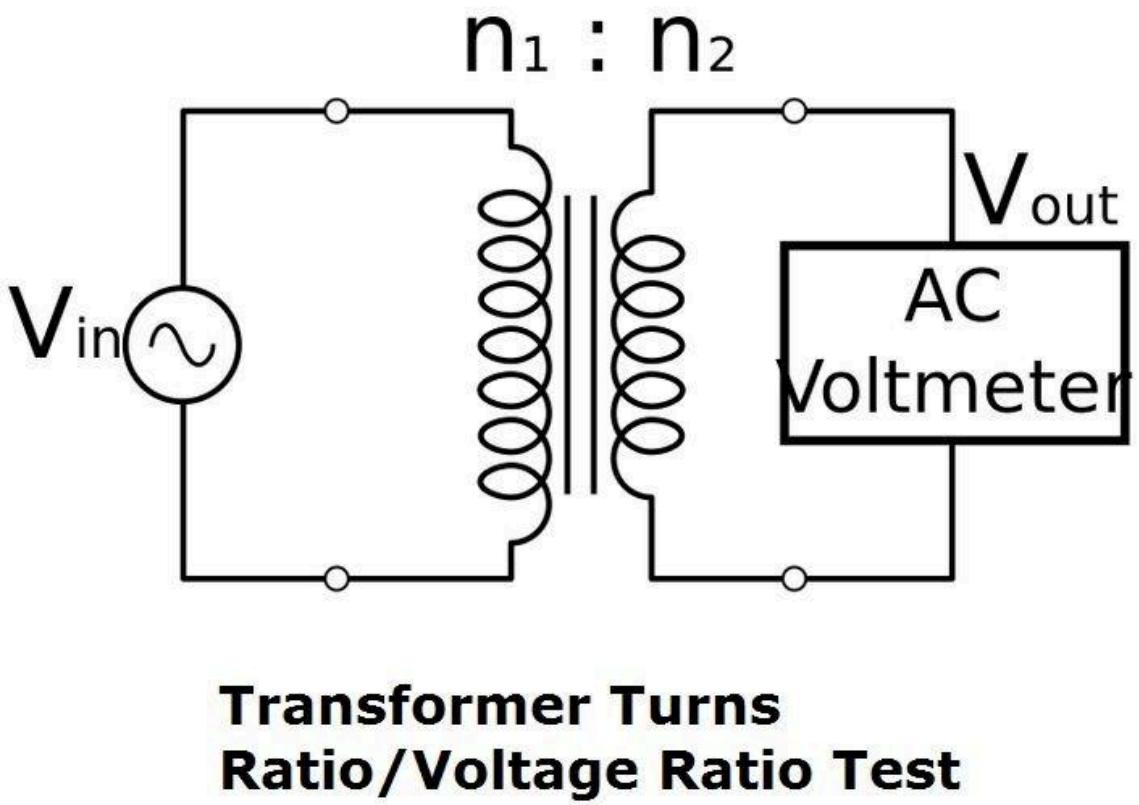


Figure 2: Circuit diagram for transformer voltage ratio measurement. [1]

1. The primary winding of the transformer was connected to a variable AC power supply.
2. Three separate secondary coils (or taps) were identified for measurement: Coil 1 (Vo1), Coil 2 (Vo2), and Coil 3 (Vo3).
3. The input voltage (Vi) was increased in steps ranging from approximately 10.6 V to 221.6 V.
4. At each step, the corresponding output voltages (Vo1, Vo2, Vo3) were recorded.

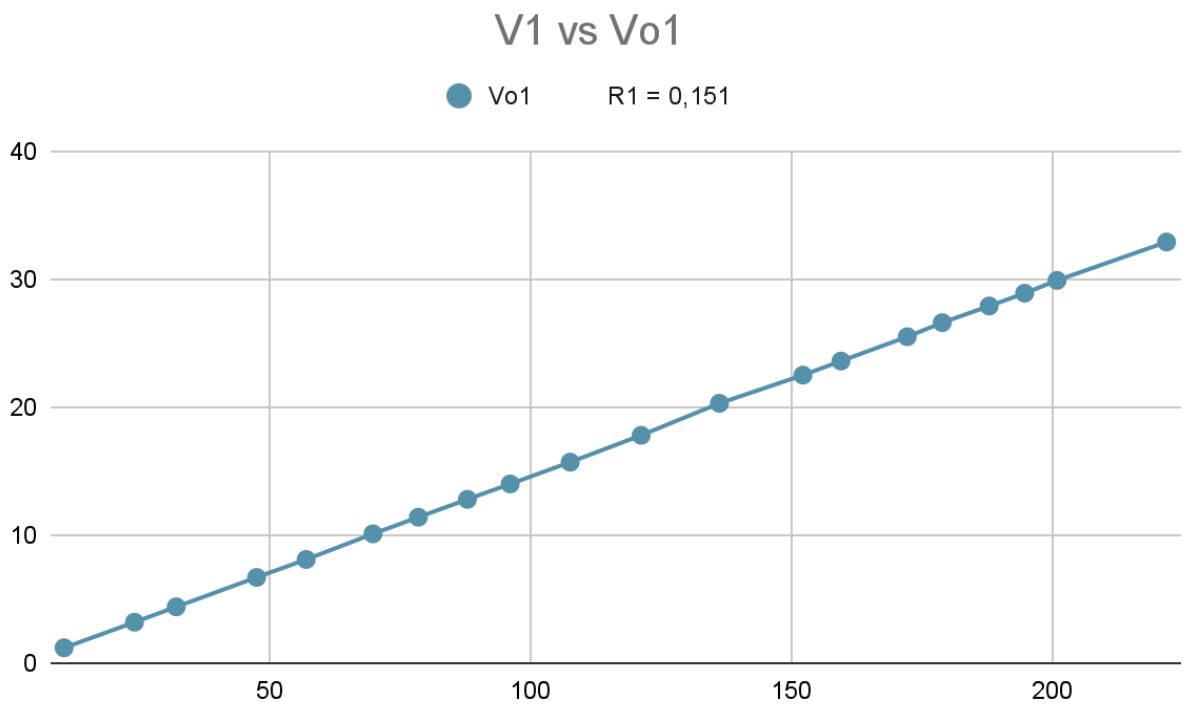
5. Data and Analysis

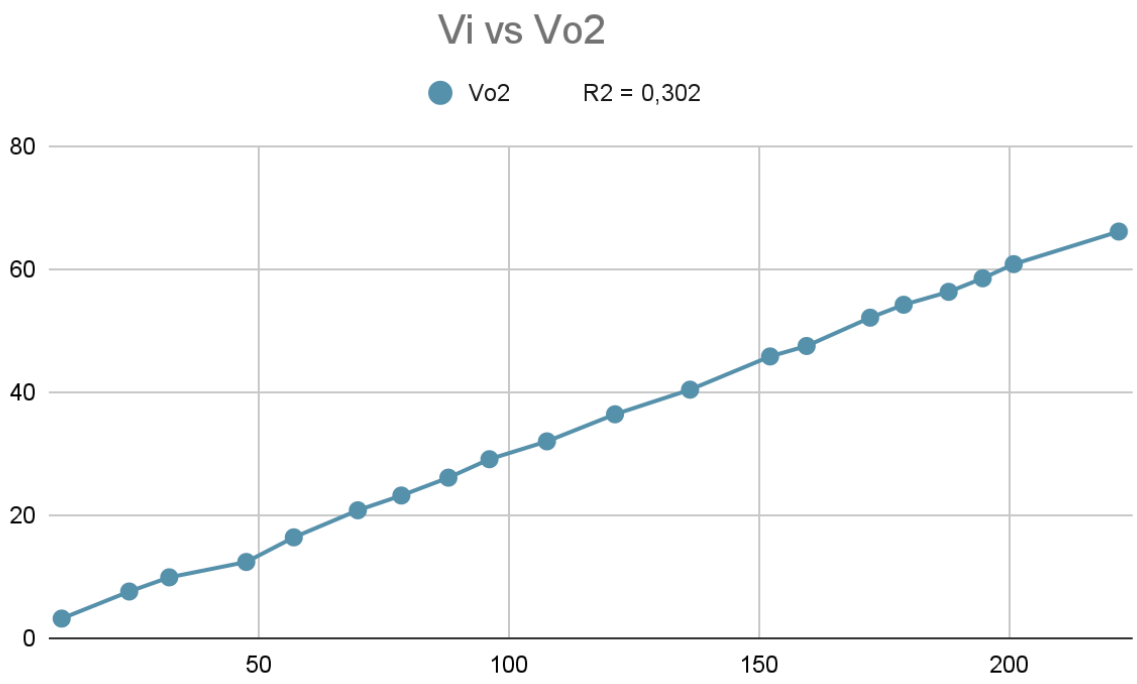
5.1 Measured Data and Graphical Analysis

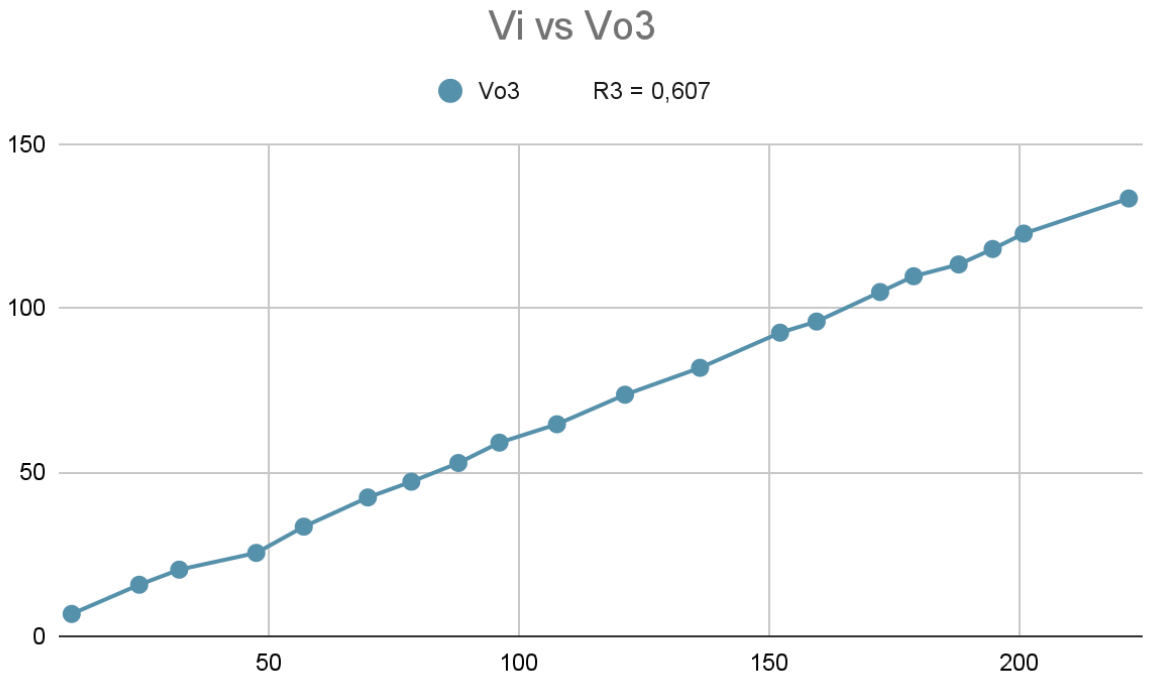
Based on the recorded experimental data, we calculated the voltage transformation ratio (k = Vo / Vi) for selected data points to observe trends.

(Sample Calculation Table)

Vi (Input)	Vo1 (Output 1)	Ratio k1 (Vo1 / Vi)
10.6 V	1.2 V	0.113
47.5 V	6.7 V	0.141
110.6 V	15.7 V	0.142
221.6 V	32.9 V	0.148


V1 vs Vo1
Vo1
R1 = 0,151
40
30
20
10
0
50
100
150
200

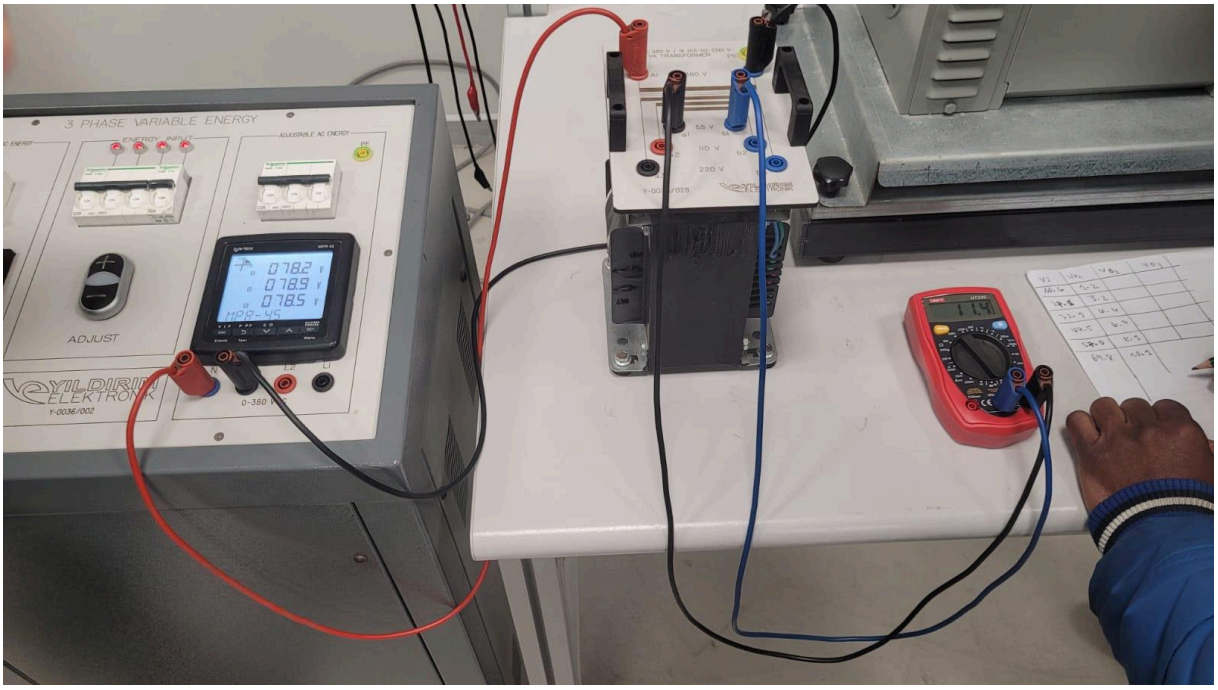

Vi vs Vo2
Vo2
R2 = 0,302
80
60
40
20
0
50
100
150
200



5.2 Experimental Measurement Photographs




3 PHASE VARIABLE ENERGY
ENERGY INPUT
ADJUSTABLE AC ENERGY
ADJUST
023.0
023.7
024.1
MOD-US
YILDIRIM ELEKTRONİK
Y-0036/002
0-380 V AC
55 V
110 V
220 V
03.21


3 PHASE VARIABLE ENERGY
ENERGY INPUT
ADJUSTABLE AC ENERGY
ADJUST
078.2
078.9
078.5
MOD-US
YILDIRIM ELEKTRONİK
Y-0036/002
55 V
110 V
220 V
1.49



6. Discussion of Results

Explanation of the variation in Vo / Vi ratios:

We observed that the ratio Vo / Vi is not perfectly constant. For Coil 1, the ratio increased from 0.113 at low voltage to 0.148 at high voltage. Ideally, this line should be flat. The variations can be attributed to the following physical factors:

1. Magnetizing Current and Impedance Drop at Low Voltages

At very low input voltages (e.g., 10.6 V), the transformer is operating at the bottom of its magnetization curve. Even with no load connected to the secondary, the primary winding draws a small "exciting current" to magnetize the core.

The primary winding has internal resistance (R1) and leakage reactance (X1). This current creates a voltage drop across the primary winding. The formula for the actual induced voltage (E1) is:

E1 = Vi - (I_exc * Z1)

Where:

- **Vi** = Input Voltage
- **I_exc** = Exciting Current
- **Z1** = Primary Impedance

At low voltages, this voltage drop is a larger percentage of the total input voltage, resulting in a lower induced voltage on the secondary side. This explains why the ratio (Vo / Vi) appears lower at the start (0.113) and "improves" or stabilizes as the voltage increases (0.148).

2. Measurement Sensitivity

The first data point for Vo1 is 1.2 V. Most standard laboratory voltmeters have lower accuracy or resolution at the very bottom of their range. The significant jump in ratio between the first and second reading suggests that measurement resolution may have contributed to the initial discrepancy.

3. Magnetic Linearity

For the mid-range voltages (approx. 50V to 150V), the ratios remain relatively stable. This indicates the "Linear Region" of the transformer where the core permeability is constant, and the device behaves closest to the ideal theoretical model.

7. Conclusion

The experiment successfully verified the transformer operation across a wide voltage range. While the device follows the approximate turns ratio Vo is proportional to Vi, deviations were observed, particularly at lower voltages. These deviations confirm that while the turns ratio is fixed physically by the number of windings, the effective voltage ratio is influenced by internal impedance voltage drops and excitation characteristics of the core.

8. Appendix

ANSWERS TO THE EVALUATION QUESTIONS

Q1. A DC supply is used because it doesn't create alternating magnetic fields in the transformer core. This avoids inductive reactance and induced voltage, meaning the measurement reflects only the actual resistance of the winding. If AC were used instead, magnetic effects and reactance would distort the reading, so you'd measure impedance, not pure resistance—leading to inaccurate results

Q2. DC voltage is applied based on the AC-rated values to control current flow and avoid overheating. Since DC lacks reactance, it allows more current to pass through. Without adjusting for this, the current could rise too high, risking damage to the windings due to overheating.

Q3. Winding resistances were calculated by applying DC voltage and measuring current. Using Ohm's law R = U / I, resistance was determined without the influence of inductive or magnetic effects—something only possible under DC conditions.

Q4. When AC voltage is applied, the measurement includes both resistance and inductive reactance—together known as impedance. AC creates magnetic flux in the transformer core, and this flux opposes current changes, adding reactance. So, the value appears higher than with DC, which measures only pure resistance.

Q5. Yes, the two methods can give different results. Ohmmeters use very small currents, minimizing heat and contact resistance effects. The R = U / I method, using higher currents, can cause slight heating in the windings and increase resistance. Contact resistance also becomes more noticeable at higher currents.

Q6. The experiment confirmed that DC measurement gives accurate transformer winding resistance by avoiding inductive effects. Under AC, reactance and magnetic behavior distort the readings, making them higher than the actual resistance. This proves why DC is essential for reliable resistance testing.

References

[1] https://eeediary.com/winding-resistance-testing-wrt-vs-transformer-turns-ratio-ttr-testing/